are singly ionized and in the ground state,[16] so that almost all of the calcium atoms are available for forming the H and K lines of calcium:

$$\begin{aligned}\left[\frac{N_1}{N_{\text{total}}}\right]_{\text{Ca II}} &\simeq \left[\frac{N_1}{N_1+N_2}\right]_{\text{Ca II}}\left[\frac{N_{\text{II}}}{N_{\text{total}}}\right]_{\text{Ca}}\\ &= \left(\frac{1}{1+[N_2/N_1]_{\text{Ca II}}}\right)\left(\frac{[N_{\text{II}}/N_{\text{I}}]_{\text{Ca}}}{1+[N_{\text{II}}/N_{\text{I}}]_{\text{Ca}}}\right)\\ &= \left(\frac{1}{1+3.79\times 10^{-3}}\right)\left(\frac{918}{1+918}\right)\\ &= 0.995.\end{aligned}$$

Now it becomes clear why the Ca II H and K lines are so much stronger in the Sun's spectrum than are the Balmer lines. There are 500,000 hydrogen atoms for every calcium atom in the solar photosphere, but only an extremely small fraction, 5.06×10^{-9}, of these hydrogen atoms are un-ionized and in the first excited state, capable of producing a Balmer line. Multiplying these two factors,

$$(500{,}000) \times (5.06 \times 10^{-9}) \approx 0.00253 = \frac{1}{395},$$

reveals that there are approximately 400 times more Ca II ions with electrons in the ground state (to produce the Ca II H and K lines) than there are neutral hydrogen atoms with electrons in the first excited state (to produce the Balmer lines). The strength of the H and K lines is *not* due to a greater abundance of calcium in the Sun. Rather, the strength of these Ca II lines reflects the sensitive temperature dependence of the atomic states of excitation and ionization.

Figure 8.11 shows how the strength of various spectral lines varies with spectral type and temperature. As the temperature changes, a smooth variation from one spectral type to the next occurs, indicating that there are only minor differences in the composition of stars, as inferred from their spectra. The first person to determine the composition of the stars and discover the dominant role of hydrogen in the universe was Cecilia Payne (1900–1979). Her 1925 Ph.D. thesis, in which she calculated the relative abundances of 18 elements in stellar atmospheres, is among the most brilliant ever done in astronomy. In Section 9.5, we will see just how the relative abundances of atoms and molecules in stellar atmospheres are measured.

8.2 ■ THE HERTZSPRUNG–RUSSELL DIAGRAM

Early in the twentieth century, as astronomers accumulated data for an increasingly large sample of stars, they became aware of the wide range of stellar luminosities and absolute magnitudes. The O stars at one end of the Harvard sequence tended to be both brighter and

[16]It is left as an exercise to show that only a very small fraction of calcium atoms are doubly ionized (Ca III).

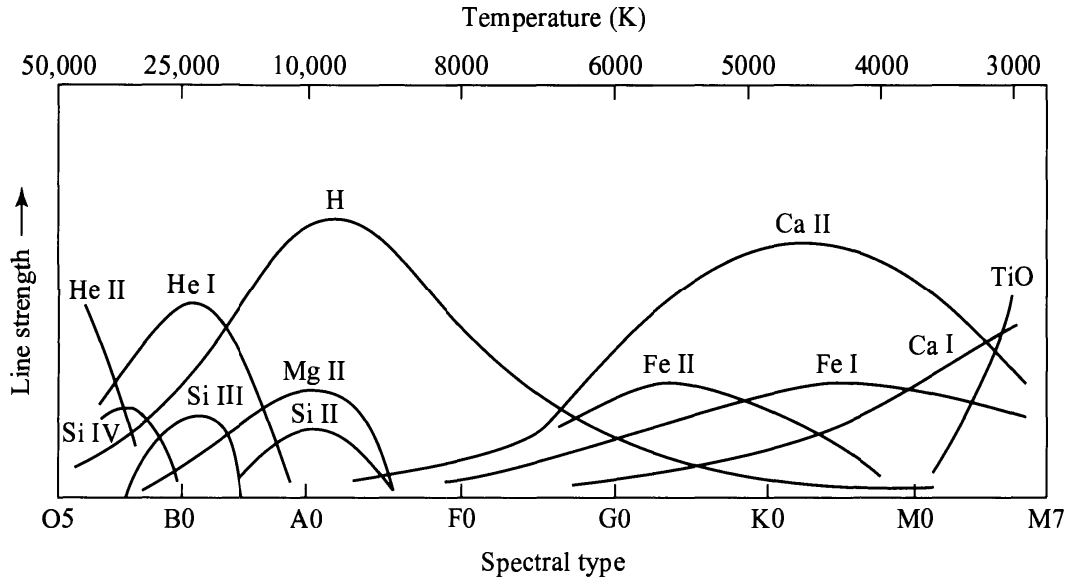

FIGURE 8.11 The dependence of spectral line strengths on temperature.

hotter than the M stars at the other end. In addition, the empirical mass–luminosity relation (see Fig. 7.7), deduced from the study of binary stars, showed that O stars are more massive than M stars. These regularities led to a theory of stellar evolution[17] that described how stars might cool off as they age. This theory (no longer accepted) held that stars begin their lives as young, hot, bright blue O stars. It was suggested that as they age, stars become less massive as they exhaust more and more of their "fuel" and that they then gradually become cooler and fainter until they fade away as old, dim red M stars. Although incorrect, a vestige of this idea remains in the terms *early* and *late* spectral types.

An Enormous Range in Stellar Radii

If this idea of stellar cooling were correct, then there should be a relation between a star's absolute magnitude and its spectral type. A Danish engineer and amateur astronomer, Ejnar Hertzsprung (1873–1967), analyzed stars whose absolute magnitudes and spectral types had been accurately determined. In 1905 he published a paper confirming the expected correlation between these quantities. However, he was puzzled by his discovery that stars of type G or later had a range of magnitudes, despite having the same spectral classification. Hertzsprung termed the brighter stars **giants**. This nomenclature was natural, since the Stefan–Boltzmann law (Eq. 3.17) shows that

$$R = \frac{1}{T_e^2}\sqrt{\frac{L}{4\pi\sigma}}. \tag{8.10}$$

If two stars have the same temperature (as inferred for stars having the same spectral type), then the more luminous star must be larger.

[17] Stellar evolution describes the change in the structure and composition of an individual star as it ages. This usage of the term *evolution* differs from that in biology, where it describes the changes that occur over generations, rather than during the lifetime of a single individual.

Hertzsprung presented his results in tabular form only. Meanwhile, at Princeton University, Henry Norris Russell (1877–1957) independently came to the same conclusions as Hertzsprung. Russell used the same term, *giant*, to describe the luminous stars of late spectral type and the term **dwarf** stars for their dim counterparts. In 1913 Russell published the diagram shown in Fig. 8.12. It records a star's observed properties: absolute magnitude on the vertical axis (with brightness increasing upward) and spectral type running horizontally (so temperature increases to the *left*). This first "Russell diagram" shows most of the features of its modern successor, the **Hertzsprung–Russell (H–R) diagram**.[18] More than 200 stars were plotted, most within a band reaching from the upper left-hand corner, home of the hot, bright O stars, to the lower right-hand corner, where the cool, dim M stars reside. This band, called the **main sequence**, contains between 80% and 90% of all stars in the H–R diagram. In the upper right-hand corner are the giant stars. A single **white dwarf**, 40 Eridani B, sits at the lower left.[19] The vertical bands of stars in Russell's diagram are a result of the discrete classification of spectral types. A more recent version of an observational H–R diagram is shown in Fig. 8.13 with the absolute visual magnitude of each star plotted versus its color index and spectral type.[20]

Figure 8.14 shows another version of the H–R diagram. Based on the average properties of main-sequence stars as listed in Appendix G, this diagram has a theorist's orientation: The luminosity and effective temperature are plotted for each star, rather than the observationally determined quantities of absolute magnitude and color index or spectral type. The uneven nature of the main sequence is an artifact of the slight differences among the references used to compile the tables in this appendix. The Sun (G2) is found on the main sequence, as is Vega (A0). Both axes are scaled logarithmically to accommodate the huge span of stellar luminosities, ranging from about $5 \times 10^{-4}\ L_\odot$ to nearly $10^6\ L_\odot$.[21] Actually, the main sequence is not a line but, rather, has a finite width, as shown in Figs. 8.12 and 8.13, owing to the changes in a star's temperature and luminosity that occur while it is on the main sequence and to slight differences in the compositions of stars. The giant stars occupy the region above the lower main sequence, with the **supergiants**, such as Betelgeuse, in the extreme upper right-hand corner. The white dwarfs (which, despite their name, are often not white at all) lie well below the main sequence.

The radius of a star can be easily determined from its position on the H–R diagram. The Stefan–Boltzmann law in the form of Eq. (8.10) shows that if two stars have the same surface temperature, but one star is 100 times more luminous than the other, then the

[18]The names of Hertzsprung and Russell were forever joined by another Danish astronomer, Bengt Strömgren (1908–1987), who suggested that the diagram be named after its two inventors. Strömgren's suggestion that star clusters be studied led to a clarification of the ideas of stellar evolution.

[19]Russell merely considered this star to be an extremely underluminous binary companion of the star 40 Eridani A; the extraordinary nature of white dwarfs was yet to be discovered. Note that the term *dwarf* refers to the stars on the main sequence and should not be confused with the *white dwarf* designation for stars lying well below the main sequence.

[20]Note that Fig. 8.13 suggests that a correlation exists between color index and spectral type, both of which are reflections of the effective temperature of the star. Recall that color index is closely related to the blackbody spectrum of a star (Section 3.6).

[21]Extremely late and early spectral types are not included in Fig. 8.14. The dimmest main-sequence stars are difficult to find, and the brightest have very short lifetimes, making their detection unlikely. As a result, only a handful of stars belonging to these classifications are known—too few to establish their average properties.

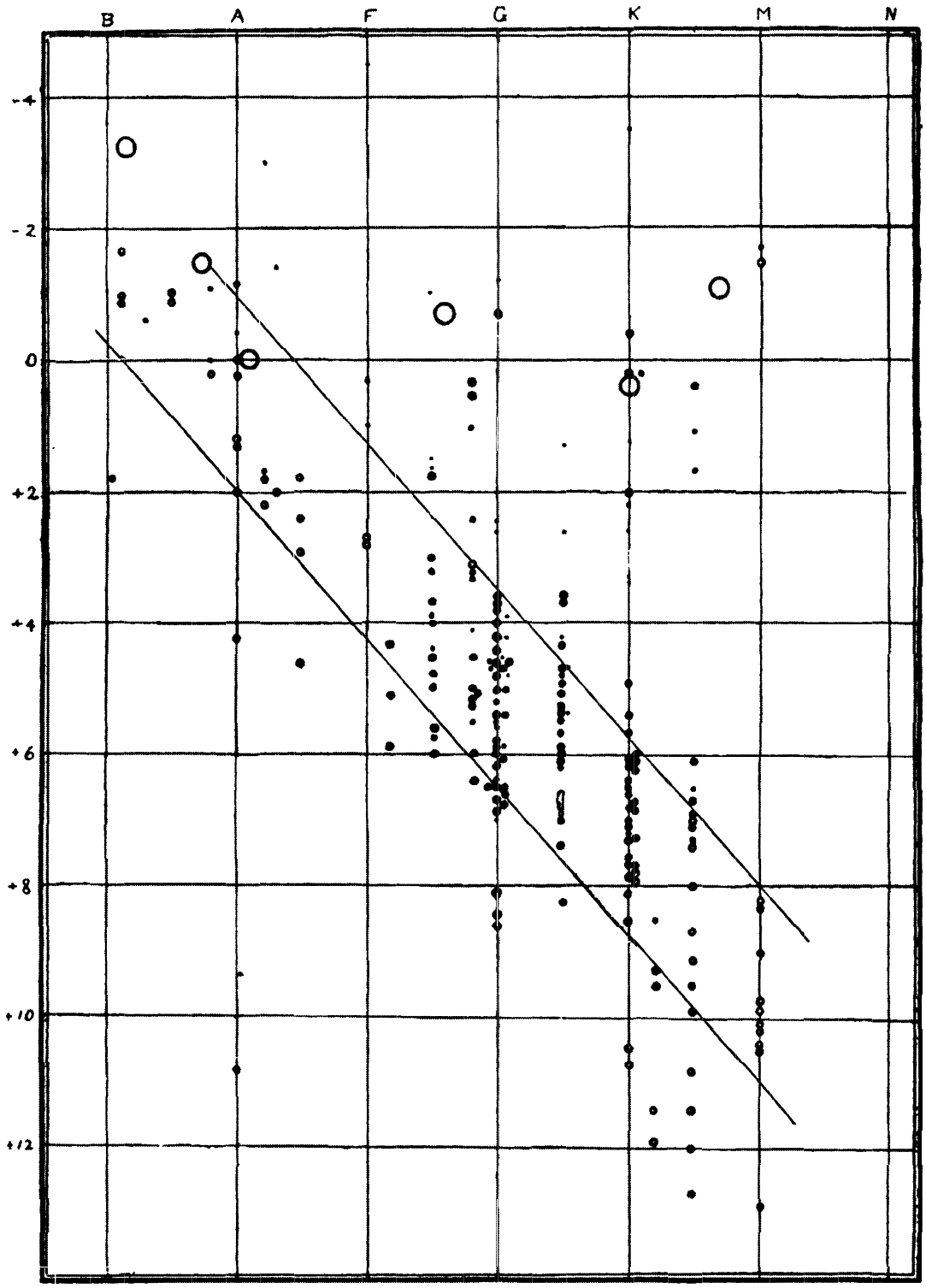

FIGURE 8.12 Henry Norris Russell's first diagram, with spectral types listed along the top and absolute magnitudes on the left-hand side. (Figure from Russell, *Nature*, *93*, 252, 1914.)

radius of the more luminous star is $\sqrt{100} = 10$ times larger. On a logarithmically plotted H–R diagram, the locations of stars having the same radii fall along diagonal lines that run roughly parallel to the main sequence (lines of constant radius are also shown in Fig. 8.14). The main-sequence stars show some variation in their sizes, ranging from roughly 20 $R_\odot$ at the extreme upper left end of the main sequence down to 0.1 $R_\odot$ at the lower right end. The giant stars fall between roughly 10 $R_\odot$ and 100 $R_\odot$. For example, Aldebaran (α Tauri),

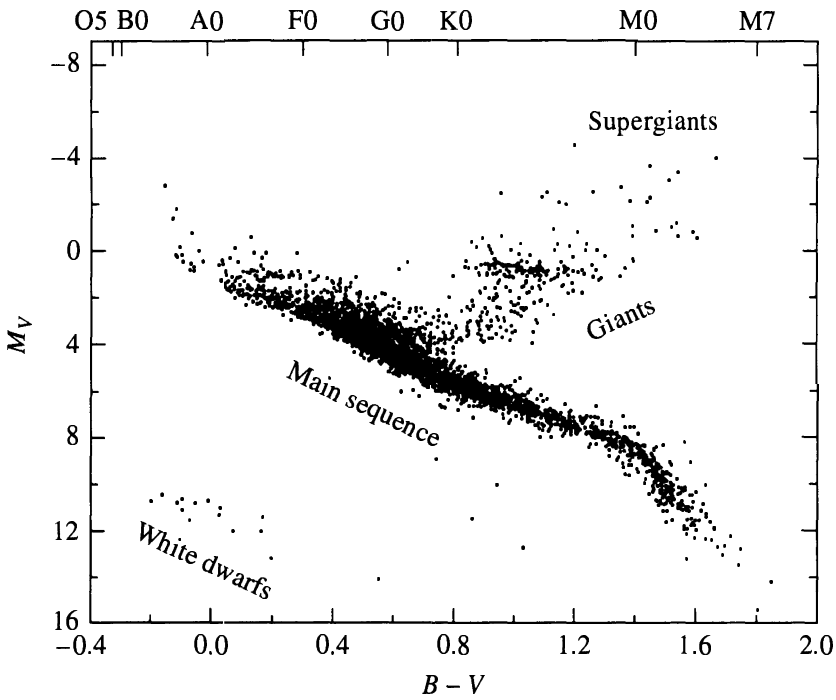

FIGURE 8.13 An observer's H–R diagram. The data are from the Hipparcos catalog. More than 3700 stars are included here with parallax measurements determined to better than 20%. (Data courtesy of the European Space Agency.)

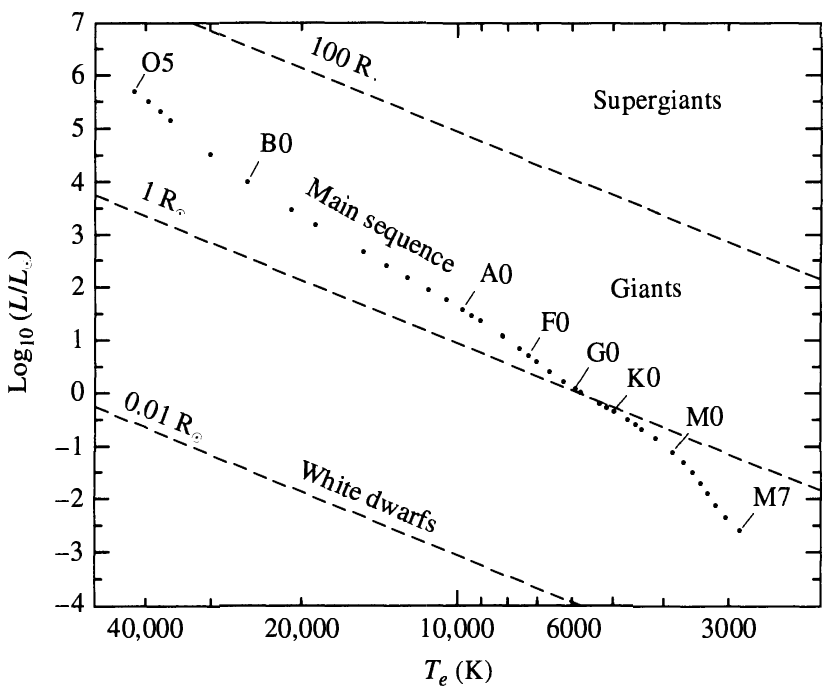

FIGURE 8.14 The theorist's Hertzsprung–Russell diagram. The dashed lines indicate lines of constant radius.

the gleaming "eye" of the constellation Taurus (the Bull), is an orange giant star that is 45 times larger than the Sun.

The supergiant stars are even larger. Betelgeuse, a pulsating variable star, contracts and expands roughly between 700 and 1000 times the radius of the Sun with a period of approximately 2070 days. If Betelgeuse were located at the Sun's position, its surface would at times extend past the orbit of Jupiter. The star μ Cephei in the constellation of Cepheus (a king of Ethiopia) is even larger and would swallow Saturn.[22]

The existence of such a simple relation between luminosity and temperature for main-sequence stars is a valuable clue that the position of a star on the main sequence is governed by a single factor. This factor is the star's *mass*.[23] The masses of stars along the main sequence are listed in Appendix G. The most massive O stars listed in that table are observed to have masses of 60 $M_\odot$,[24] and the lower end of the main sequence is bounded by M stars having at least 0.08 $M_\odot$.[25] Combining the radii and masses known for main-sequence stars, we can calculate the average density of the stars. The result, perhaps surprising, is that main-sequence stars have roughly the same density as water. Moving up the main sequence, we find that the larger, more massive, early-type stars have a *lower* average density.

Example 8.2.1. The Sun, a G2 main-sequence star, has a mass of $M_\odot = 1.9891 \times 10^{30}$ kg and a radius of $R_\odot = 6.95508 \times 10^8$ m. Its average density is thus

$$\overline{\rho}_\odot = \frac{M_\odot}{\frac{4}{3}\pi R_\odot^3} = 1410 \text{ kg m}^{-3}.$$

Sirius, the brightest-appearing star in the sky, is classified as an A1 main sequence star with a mass of 2.2 $M_\odot$ and a radius of 1.6 $R_\odot$. The average density of Sirius is

$$\overline{\rho} = \frac{2.2 \text{ M}_\odot}{\frac{4}{3}\pi(1.6 \text{ R}_\odot)^3} = 760 \text{ kg m}^{-3} = 0.54\,\overline{\rho}_\odot,$$

which is about 76 percent of the density of water. However, this is enormously dense compared to a giant or supergiant star. The mass of Betelgeuse is estimated to lie between 10 and 15 $M_\odot$; we will adopt 10 $M_\odot$ here. For illustration, if we take the maximum radius of this pulsating star to be about 1000 $R_\odot$, then the average density of Betelgeuse (at maximum size) is roughly

$$\overline{\rho} = \frac{10 \text{ M}_\odot}{\frac{4}{3}\pi(1000 \text{ R}_\odot)^3} = 10^{-8}\,\overline{\rho}_\odot!$$

Thus Betelgeuse is a tenuous, ghostly object—a hundred thousand times less dense than the air we breathe. It is difficult even to define what is meant by the "surface" of such a wraith-like star.

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[22] μ Cephei is a pulsating variable like Betelgeuse and has a period of 730 days. One of the reddest stars visible in the night sky, μ Cephei, is known as the *Garnet Star*.

[23] In Chapter 10 we will see how mass determines the location of a star on the main sequence.

[24] Theoretical calculations indicate that main-sequence stars as massive as 90 $M_\odot$ may exist, and recent observations have been made of a few stars with masses estimated near 100 $M_\odot$; see Section 15.3.

[25] Stars less massive than 0.08 $M_\odot$ have insufficient temperatures in their cores to support significant nuclear burning (see Chapter 10).

Morgan–Keenan Luminosity Classes

Hertzsprung wondered whether there might be some difference in the spectra of giant and main-sequence stars of the same spectral type (or same effective temperature). He found just such a variation in spectra among the stars cataloged by Antonia Maury. In her classification scheme she had noted line width variations that she referred to as a *c*-characteristic. The subtle differences in the relative strengths of spectral lines for stars of similar effective temperatures and different luminosities are depicted in Fig. 8.15. The work begun by Hertzsprung and Maury, and further developed by other astronomers, culminated in the 1943 publication of the *Atlas of Stellar Spectra* by William W. Morgan (1906–1994) and Phillip C. Keenan (1908–2000) of Yerkes Observatory. Their atlas consists of 55 prints of spectra that clearly display the effect of temperature and luminosity on stellar spectra and includes the criteria for the classification of each spectrum. The *MKK Atlas* established the *two-dimensional* Morgan–Keenan (M–K) system of spectral classification.[26] A **luminosity class**, designated by a Roman numeral, is appended to a star's Harvard spectral type. The numeral "I" (subdivided into classes Ia and Ib) is reserved for the supergiant stars, and "V" denotes a main-sequence star. The ratio of the strengths of two closely spaced lines is often employed to place a star in the appropriate luminosity class. In general, for stars of the same spectral type, narrower lines are usually produced by more luminous stars.[27] The Sun is a G2 V star, and Betelgeuse is classified as M2 Ia.[28] The series of Roman numerals extends below the main sequence; the subdwarfs (class VI or "sd") reside slightly to the left of the main sequence because they are deficient in metals. The M–K system does not extend to the white dwarfs, which are classified by the letter D. Figure 8.16 shows the corresponding divisions on the H–R diagram and the locations of a selection of specific stars, and Table 8.3 lists the luminosity classes.

The two-dimensional M–K classification scheme enables astronomers to locate a star's position on the Hertzsprung–Russell diagram based *entirely* on the appearance of its spectrum. Once the star's absolute magnitude, M, has been read from the vertical axis of the H–R diagram, the *distance* to the star can be calculated from its apparent magnitude, m, via Eq. (3.5),

$$d = 10^{(m-M+5)/5},$$

where d is in units of parsecs. This method of distance determination, called **spectroscopic parallax**, is responsible for many of the distances measured for stars,[29] but its accuracy is limited because there is not a perfect correlation between stellar absolute magnitudes and luminosity classes. The intrinsic scatter of roughly ± 1 magnitude for a specific luminosity class renders d uncertain by a factor of about $10^{1/5} = 1.6$.

[26] Edith Kellman of Yerkes printed the 55 spectra and was a co-author of the atlas; hence the additional "K" in *MKK Atlas*.

[27] In Section 9.5, we will find that because the atmospheres of more luminous stars are less dense, there are fewer collisions between atoms. Collisions can distort the energies of atomic orbitals, leading to broadening of the spectral lines.

[28] Betelgeuse, a pulsating variable star, is sometimes given the intermediate classification M2 Iab.

[29] Since the technique of parallax is not involved, the term *spectroscopic parallax* is a misnomer, although the name does at least imply a distance determination.

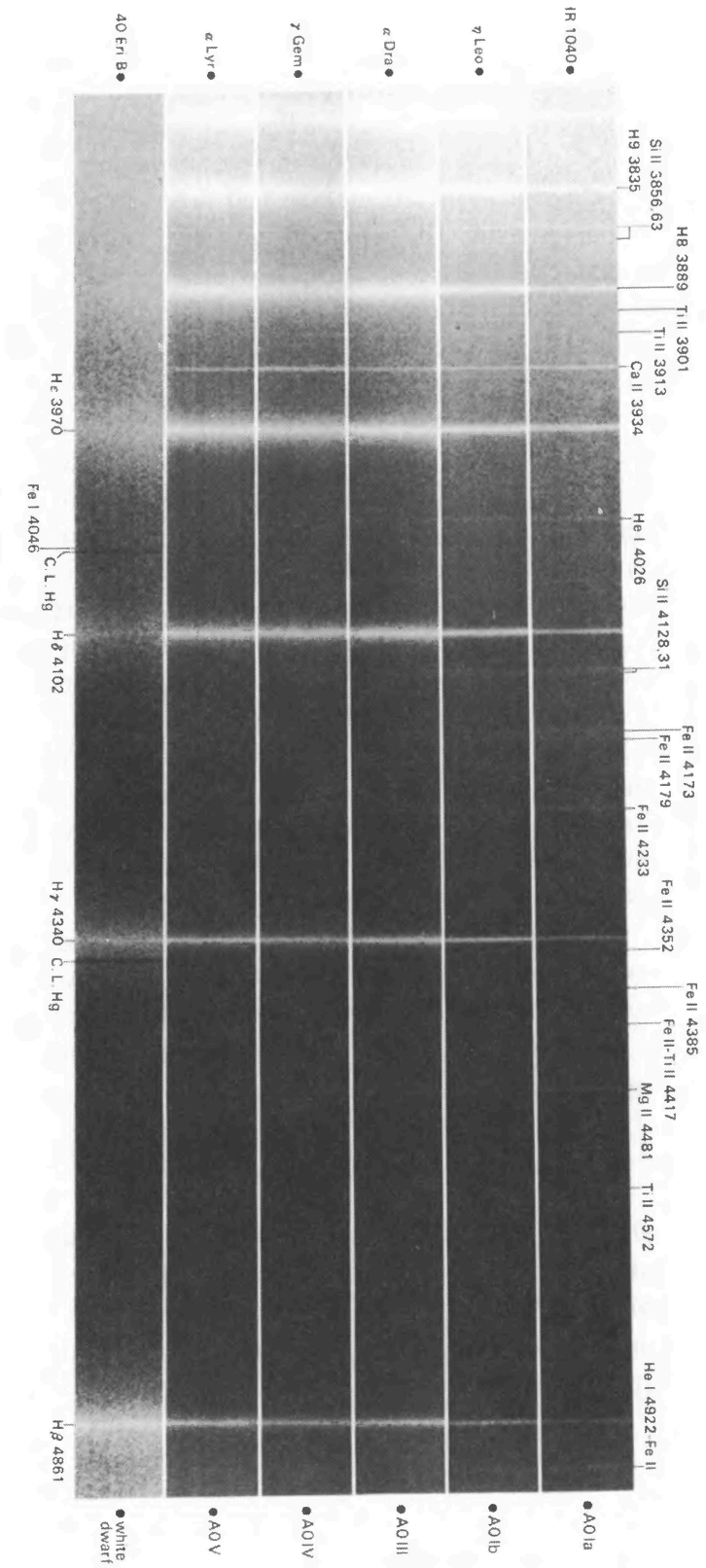

FIGURE 8.15 A comparison of the strengths of the hydrogen Balmer lines in types A0 Ia, A0 Ib, A0 III, A0 IV, A0 V, and a white dwarf, showing the narrower lines found in supergiants. These spectra are displayed as negatives, so absorption lines appear bright. (Figure from Yamashita, Nariai, and Norimoto, *An Atlas of Representative Stellar Spectra*, University of Tokyo Press, Tokyo, 1978.)

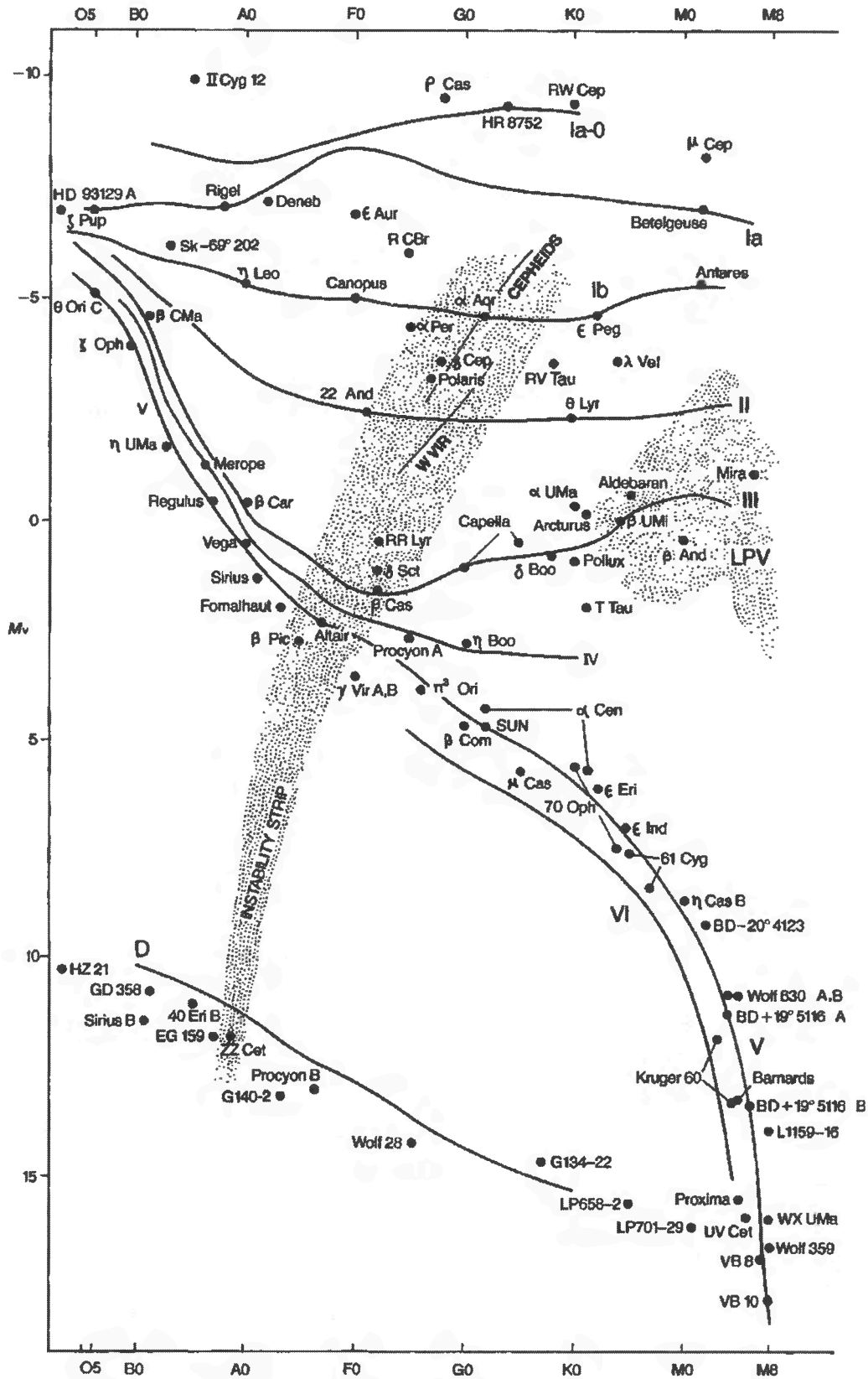

FIGURE 8.16 Luminosity classes on the H–R diagram. (Figure from Kaler, *Stars and Stellar Spectra*, © Cambridge University Press 1989. Reprinted with the permission of Cambridge University Press.)

TABLE 8.3 Morgan–Keenan Luminosity Classes.

Class	Type of Star
Ia-O	Extreme, luminous supergiants
Ia	Luminous supergiants
Ib	Less luminous supergiants
II	Bright giants
III	Normal giants
IV	Subgiants
V	Main-sequence (dwarf) stars
VI, sd	Subdwarfs
D	White dwarfs

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